Component List

AI + IoT Smart Home Security System with Plant Care and Mold-Risk Alert. Prices in Turkish Lira (TL), VAT included. Already have means the team owns it, so it costs nothing. A ~ means an estimate; other prices were taken from Turkish shops (Direnc.net, Robotistan). Stock and prices change, so get quotes before buying.

A. Brains and communication

Component	What it is	What it does / feature	Qty	Price (TL)
ESP32-S3 LilyGO T-Display	A Wi-Fi computer on a chip with a built-in screen.	Main controller: door lock, fingerprint, keypad, and shows status on its own screen.	1	Already have
ESP32-C3 Super Mini	A tiny Wi-Fi computer on one chip.	Second controller: plant care, bathroom-corner sensors, clap light.	1	Already have
ESP32-CAM	An ESP32 with a small camera.	Door/room camera; feeds the AI (people, movement, fire).	1	941
Team laptop	The computer you already have.	Runs the AI, server, dashboard and Jarvis.	1	Already have

B. Security and door access

Component	What it is	What it does / feature	Qty	Price (TL)
AS608 fingerprint sensor	A small fingerprint reader.	Unlocks the door for enrolled fingers.	1	814
Keypad 4x4	A 16-button number pad.	Unlocks the door with a password.	1	~80
Solenoid lock	A small electric bolt.	The door's lock.	1	256
Relay boards	Electric switches controlled by a tiny signal.	Safely switch the lock, light, pump and humidifier.	2	148
Buzzer	A small beeping speaker.	Alarm for intruders, fire and mold risk.	1	~25
Status display	A small screen.	Shows armed / disarmed and status (T-Display's screen, plus your spare display).	1	Already have
Face ID (software)	Face recognition on the laptop.	Unlocks the door; builds the 'who is here' list.	-	0

C. AI vision (free software)

Component	What it is	What it does / feature	Qty	Price (TL)
YOLO model	AI that finds objects in video.	Detects people, movement, fire and smoke.	-	0
Face recognition library	AI that compares faces to saved ones.	Tells known people from strangers.	-	0

D. Smart control

Component	What it is	What it does / feature	Qty	Price (TL)
MAX9814 microphone	A sensitive mic with amplifier.	Hears claps to switch the light.	1	192
Jarvis voice assistant	Free speech-to-text, AI brain and voice.	Talk to the house; it answers and acts.	-	0

E. Environment and fire sensors

Component	What it is	What it does / feature	Qty	Price (TL)
BME680 sensor	Measures temperature, humidity, pressure and air quality.	Watches the bathroom corner for mold-friendly conditions.	1	Already have
MQ-2 gas sensor	Smells smoke and gas.	Second fire / gas warning.	1	70
Flame sensor	Sees the infrared light of a flame.	Fast fire warning.	1	27

F. Plant care

Component	What it is	What it does / feature	Qty	Price (TL)
Soil moisture sensors	Probes that read how wet soil is.	Tell the system a plant is thirsty.	2	101
Water pump 12V	A small pump.	Waters the plants automatically.	1	581
LED grow light strip	A light for plants.	Gives plants light on a timer.	1	~300
Plants, soil, pots	The plants themselves.	What the system looks after.	-	~200

G. Bathroom corner (mold-risk simulation)

Component	What it is	What it does / feature	Qty	Price (TL)
Small plastic tub or house corner	A small enclosed space.	The 'bathroom' where damp, warm conditions are created.	1	~150
Mini humidifier	A small mist maker.	Raises humidity to the mold-friendly level for the demo.	1	~250
Heater / heat mat (optional)	A small warmer.	Adds the warmth that mold likes.	1	~300
Small fan	A small fan.	Switches on as the response to a mold-risk alert.	1	~50

H. Power, wiring and model house

Component	What it is	What it does / feature	Qty	Price (TL)
12V and 5V adapters	Wall chargers.	Power the lock, pump and boards.	2	~400
Jumper wires	Connecting wires.	Connect everything together.	-	Already have
Resistors, MOSFET, terminals, perfboard	Basic circuit parts.	Finish the circuits.	-	~250
Model house materials	Foam board / MDF / acrylic, glue.	The demo house with door, room, camera spot and bathroom corner.	1	~700

Total cost (to buy)

Section	TL
A. Brains and communication (ESP32 boards already owned; 1 camera to buy)	941
B. Security and door access	1,323
C. AI vision	0
D. Smart control	192
E. Environment and fire sensors (BME680 already owned)	97
F. Plant care	1,182
G. Bathroom corner	750
H. Power, wiring and model house	1,350
GRAND TOTAL	**about 5,800**

This is well under the 7,000-10,000 TL budget. A second ESP32-CAM (about 941 TL) would bring the total to about 6,800 TL.

What the project is

A smart home security system built on a small model house. A camera and AI watch for people, movement and fire; the door opens only for a recognised fingerprint, face or password. The house also looks after plants and warns you when a damp, warm area (like a bathroom) reaches the conditions where mold grows.

All features in plain words

Intruder detection	A camera watches the door and room. AI tells a person from a pet or object and raises an alarm.
Fire and smoke detection	The camera AI, a flame sensor and a gas sensor all watch for fire.
Smart door lock	Opens with a fingerprint, a face scan or a password.
Who is home	Face recognition shows who is present, who is away, and flags strangers.
Clap light switch	Clap and the light turns on or off.
Jarvis voice assistant	Talk to the house: ask for status or give commands.
Plant care	Sensors check the soil; the pump waters and the light runs on a timer.
Mold-risk alert	A bathroom corner is made damp and warm, the perfect conditions for mold. The sensor spots it and raises an alert, switches on a fan and warns you.
Dashboard and alerts	One web page shows the camera, sensors, history and alerts, and can notify your phone.